

TIKANLIK, China

WMO: 517650

Lat: 40.6475N

Lon: 87.6943E

Elev: 2779

StdP: 13.28

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	2.8	-10.7	3.4	14.4	-7.1	4.2	15.8	12.1	26.6	10.6	26.1	1.3	250	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.7	100.4	66.5	98.0	65.9	95.6	65.1	71.6	91.5	69.3	90.1	67.6	89.0	4.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	103.2	81.7	62.7	94.4	78.8	60.3	86.8	78.1	37.4	91.3	35.3	90.3	33.7	89.0	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
16.3	13.4	11.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-3.3	105.4	2.8	2.1	-5.3	106.9	-7.0	108.1	-8.5	109.3	-10.5	110.8
			-4.0	75.0	2.7	3.5	-6.0	77.5	-7.6	79.6	-9.1	81.5	-11.1	84.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	18.1	30.0	46.1	60.7	70.8	79.1	82.1	79.6	69.2	53.8	37.5	21.7	(6)
(7)		DBStd	23.05	5.37	7.09	8.10	7.77	6.56	5.22	4.56	4.63	5.62	6.80	6.90	6.21	(7)
(8)		HDD50	3028	990	560	174	12	0	0	0	0	40	376	876	(8)	
(9)		HDD65	5758	1455	980	586	171	25	1	0	23	351	825	1341	(9)	
(10)		CDD50	4549	0	0	53	332	645	872	994	918	577	157	1	0	(10)
(11)		CDD65	1804	0	0	0	41	205	423	529	453	150	3	0	0	(11)
(12)		CDH74	23258	0	0	15	692	2733	5378	6843	5652	1802	143	0	0	(12)
(13)		CDH80	12138	0	0	0	195	1216	2944	3949	3126	688	20	0	0	(13)
(14)	Wind	WSAvg	4.0	2.5	3.2	4.6	5.7	5.8	5.0	4.4	4.0	3.9	3.3	2.9	2.6	(14)
(15)	Precipitation	PrecAvg	1.4	0.0	0.0	0.0	0.1	0.1	0.3	0.4	0.2	0.1	0.0	0.0	0.0	(15)
(16)		PrecMax	3.0	0.2	0.0	0.6	0.6	1.8	1.5	1.6	1.4	0.7	0.4	0.4	0.1	(16)
(17)		PrecMin	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.8	0.0	0.0	0.1	0.2	0.3	0.4	0.4	0.3	0.2	0.1	0.1	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	40.5	55.7	76.0	90.4	96.8	101.4	104.5	102.4	94.7	82.7	65.0	44.3	(19)	
(20)		MCWB	30.1	39.4	49.7	59.0	61.7	65.3	67.6	67.3	63.4	56.3	45.3	33.8	(20)	
(21)		DB	36.0	51.4	71.5	85.4	93.3	98.5	101.4	99.1	90.4	77.9	60.4	39.7	(21)	
(22)		MCWB	27.6	37.4	47.6	55.5	60.1	64.8	67.5	66.2	61.7	54.0	43.4	30.9	(22)	
(23)		DB	32.7	48.2	67.7	81.4	90.2	96.0	98.6	96.4	87.3	74.0	56.3	36.1	(23)	
(24)		MCWB	25.5	35.4	45.8	53.7	59.0	64.2	67.0	65.7	60.7	52.0	41.6	28.7	(24)	
(25)		DB	29.6	44.4	63.3	77.8	87.1	93.6	95.9	93.7	84.0	69.8	51.4	32.6	(25)	
(26)		MCWB	23.5	33.0	43.7	52.0	57.7	63.5	66.4	65.3	59.7	50.5	39.1	26.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	30.3	40.0	50.9	59.4	66.1	72.6	75.1	74.4	69.1	57.5	46.1	34.5	(27)	
(28)		MCDWB	40.0	54.3	73.4	87.9	89.2	91.3	94.0	90.8	84.7	77.0	63.2	43.0	(28)	
(29)		WB	27.8	37.8	48.1	56.5	62.0	68.6	72.1	71.0	65.2	55.6	43.7	31.3	(29)	
(30)		MCDWB	35.5	50.8	70.4	83.0	88.4	90.1	92.6	90.5	84.0	75.4	59.1	39.0	(30)	
(31)		WB	25.7	35.7	46.0	54.4	60.0	66.4	69.8	68.8	62.8	53.4	41.8	28.9	(31)	
(32)		MCDWB	32.3	47.7	66.8	79.4	86.9	89.3	91.1	89.3	82.3	71.6	55.7	35.5	(32)	
(33)		WB	23.5	33.2	43.9	52.5	58.4	64.7	68.0	67.1	60.9	51.2	39.5	26.7	(33)	
(34)		MCDWB	29.3	44.0	62.8	75.9	84.0	88.3	89.8	88.2	80.7	67.9	51.1	32.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	24.8	27.1	27.8	27.2	26.9	25.0	24.7	27.0	29.5	31.5	26.8	21.9	(35)	
(36)		5% DB	27.4	29.5	31.9	32.2	32.3	29.5	30.0	31.5	33.3	35.1	32.4	24.4	(36)	
(37)		MCWBR	20.2	18.2	16.2	13.8	12.4	11.4	11.3	12.3	13.5	15.4	18.0	16.7	(37)	
(38)		5% WB	26.8	28.9	31.5	29.9	28.8	25.9	25.5	28.0	30.5	33.7	31.9	23.5	(38)	
(39)		MCWBR	20.0	18.0	16.1	13.1	12.1	11.7	12.1	13.8	13.9	14.7	18.0	16.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.353	0.416	0.498	0.566	0.587	0.613	0.587	0.579	0.588	0.488	0.400	0.358	(40)	
(41)		taud	2.040	1.900	1.717	1.612	1.620	1.621	1.697	1.698	1.634	1.768	1.949	2.017	(41)	
(42)		Ebn at Noon	242	241	234	228	228	222	226	223	207	213	219	226	(42)	
(43)		Edh at Noon	37	50	67	80	81	81	75	72	72	56	41	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	822	1141	1527	1859	2167	2237	2161	2008	1737	1328	934	701	(44)	
(45)		RadStd	56	41	57	98	94	126	78	97	58	43	38	68	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.72	N/A	N/A	+0.72	-1.96	-3.06	N/A	-198	+189	+104	(46)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	-1.15	N/A	N/A	N/A	+142	N/A	(47)

Nomenclature: See separate page